

CRC Error Detection — Presentation Quick Revision

A simple revision sheet to read before your CRC presentation.

1. What is CRC?

CRC means **Cyclic Redundancy Check**. It is an error-detection technique used to check whether data was changed or corrupted during transmission.

Important: CRC detects an error; it does not correct the error.

2. Why do we need CRC?

When data is sent from one computer to another, a bit can accidentally change during transmission.

Correct: 1101011011

Changed: 1101111011

CRC lets the receiver check whether the received data is still correct.

3. The two inputs

Input	Example	Purpose
Data	1101011011	Original binary data.
Generator	10011	Binary pattern used for the CRC calculation.

4. Why do we add zeros?

The generator **10011** has 5 bits, so we add **5 – 1 = 4 zeros** to the data to make space for the CRC remainder.

1101011011 → 11010110110000

5. XOR — the main operation

CRC uses modulo-2 arithmetic, so it uses XOR instead of normal addition. There are no carries or borrows.

XOR	Result
0 XOR 0	0
0 XOR 1	1
1 XOR 0	1
1 XOR 1	0

Example:

11010

10011

01001

6. Our example

Data: 1101011011 **Generator:** 10011 **CRC:** 1110

After repeated XOR division, the final remainder is **1110**.

1101011011 + 1110 = **11010110111110**

This final sequence is the **transmitted frame**.

7. What happens at the receiver?

The receiver checks the received frame again using the **same generator**.

Remainder = 0000 → NO ERROR DETECTED

Remainder ≠ 0000 → ERROR DETECTED

8. What your application buttons mean

Button / Option	Meaning
Generate CRC	Calculates the CRC remainder.
Copy Transmitted Frame	Copies the correct frame for checking.
No Error	Keeps the frame unchanged.
Single Bit Error	Deliberately changes one bit.
Multiple Bit Error	Deliberately changes several bits.
Random Error	Changes bits at random positions.
Verify CRC	Acts like the receiver and checks the frame.

9. The whole application in one flow

```
SENDER
↓
Data: 1101011011
↓
Calculate CRC
↓
CRC: 1110
↓
Transmitted Frame: 11010110111110
↓
CHANNEL
↓
(an error may or may not occur)
↓
RECEIVER
↓
Check using the same generator
↓
0000 → NO ERROR DETECTED
Non-zero → ERROR DETECTED
```

10. What to say during the presentation

“CRC stands for Cyclic Redundancy Check. It is an error detection technique used to check whether data has been corrupted during transmission. In my application, the sender enters binary data and a generator. The application performs modulo-2 division using XOR and calculates the CRC remainder. The CRC is attached to the original data to form the transmitted frame. At the receiver side, the same generator is used to verify the received frame. If the remainder is zero, no error is detected. If the remainder is non-zero, an error is detected. The application can also simulate single-bit, multiple-bit and random errors.”

11. Easy demo order

1. Enter Data and Generator.
2. Click Generate CRC.
3. Explain the CRC remainder.
4. Show the Transmitted Frame.
5. Copy/send the frame.
6. Verify with No Error → show zero remainder.
7. Introduce a Single Bit Error.
8. Verify again → show that an error is detected.
9. If there is time, demonstrate Multiple Bit or Random Error.

12. Quick viva answers

What is CRC? — Cyclic Redundancy Check; an error-detection technique.

Does CRC correct errors? — No. It detects errors but does not correct them.

Why XOR? — CRC uses modulo-2 arithmetic, where XOR replaces normal addition.

Why append zeros? — To make space for the CRC remainder.

What is the generator? — The binary divisor/pattern used in the CRC calculation.

What does zero remainder mean? — No error is detected.

What does non-zero remainder mean? — An error is detected.

■ Remember: Data → Add zeros → XOR division → CRC → Attach CRC → Transmit → Check again → Zero = No error, Non-zero = Error.